Characteristic Drift Spectra of Complex Systems

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***Abstract*—When a complex system is observed repeatedly, an apparent change in the data can arise for two fundamentally different reasons: we may simply have drawn a new finite sample from the same underlying organization, or the organization generating the observations may itself have changed. Distinguishing these cases is the central problem of drift measurement. We introduce a *characteristic drift spectrum* designed for this purpose. For repeated samples from the same underlying organization, the estimated spectrum should converge toward the same population fingerprint as sample size grows; a change that alters the system's conditional, residual, or informational organization instead produces a persistent spectral displacement. The construction begins from a generic learning objective: estimate, for each coordinate, its conditional distribution given the remainder of the state, and use these learned conditionals to expose progressively lower-persistence residual populations. At every adjacent level we compute three nonredundant probes—structural drift, conditional information drift, and compression-based Kolmogorov drift—because reconstruction geometry, conditional description length, and model-independent compressibility can change separately. Their ordered sequence across depth forms a task-free, multiscale fingerprint of systemic organization. In General Social Survey data, the spectrum sharply detects the known data-collection regime change beginning in 2021 and then moves partway back toward the pre-2021 regime under the mixed-mode designs of 2022 and 2024.**

Why a characteristic drift spectrum?

Repeated observations of a complex system are never identical, even when the system itself has not changed. The central problem is therefore not merely to decide whether two finite datasets are different, but whether the difference reflects ordinary re-realization of the same underlying organization or a genuine change in that organization. Two independent samples from the same unchanged system will differ simply because they are different realizations. A system-level drift representation should therefore be *stable to re-realization but sensitive to reorganization*. Let $\mathcal{S}(P)$ denote the population spectrum induced by an underlying distribution P, and let $\widehat{\mathcal{S}}_n(P)$ be its estimate from n observations. Under stable, consistent conditional learning and empirical estimation, two independent realizations from the same system satisfy

$$d\Big(\widehat{\mathcal{S}}_n(P), \widehat{\mathcal{S}}_m(P)\Big) \longrightarrow 0, \qquad n, m \to \infty. \tag{1}$$

Sampling variation therefore contracts with increasing data. If the underlying system changes from P to Q in a way that alters the conditional organization, the induced residual hierarchy, or the information structure interrogated by the spectrum, then $\mathcal{S}(P)$ and $\mathcal{S}(Q)$ need not coincide; the resulting displacement persists rather than vanishing as a finite-sample effect. This is the defining use of the spectrum: repeated realizations of the same organization approach the same fingerprint, whereas organizational drift deforms that fingerprint.

Classical dataset- and concept-drift formulations are commonly tied to a specified prediction problem, a response variable, or a global comparison between distributions [1–3]. Classifier two-sample tests make this explicit by learning a discriminator between samples and using its performance as evidence of distributional change [4]. These are appropriate when the scientific object is a task. Here the scientific object is the system itself: there may be no privileged Y, and the relevant change may lie in the network of conditional relations organizing the state space. The corresponding learning objective is therefore to recover the family of full conditionals $\{P(X_i \mid X_{-i})\}_{i=1}^{p}$, rather than to optimize a single selected response. The spectrum is built from this conditional-learning objective and is agnostic to the particular algorithm used to realize it. We therefore seek a response-variable-free, multiscale fingerprint of organization rather than a single discrimination score.

The spectrum is not claimed to be an injective encoding of the complete joint distribution: distinct distributions can in principle agree on a finite collection of probes. Its purpose is narrower and operational—to make persistent changes in the aspects of organization interrogated by those probes separable from finite-sample re-realization. Structural depth then records *where* such change appears in the hierarchy of system states.

Conditional organization and the intrinsic residual hierarchy

The construction requires a learning procedure that estimates the conditional organization of a multivariate system, not a particular modeling algorithm. For categorical state $X = (X_1, \ldots, X_p)$ and row population R, let

$$M(R) = \{\phi_i(\cdot \mid x_{-i})\}_{i \in T_M} \tag{2}$$

denote a collection of target-wise conditional predictors, where ϕ_i estimates the distribution of X_i given the remaining coordinates X_{-i}. Any sufficiently accurate and stable conditional learner can in principle instantiate the spectrum. In the experiments reported here we realize this learning objective with Large Science Models (LSMs) [5,6], which at this level of abstraction provide precisely such a collection of target-wise conditional models. The drift construction itself is therefore not specific to the LSM algorithm.

For row x, define mean conditional log-persistence

$$P_M(x) = \frac{1}{n_M(x)} \sum_{i \in T_M(x)} \log_2[\max\{\phi_i(x_i \mid x_{-i}), \varepsilon\}]\,. \tag{3}$$

$P_M(x)$ is the average log support assigned to the observed coordinates when each coordinate is predicted from the remainder of the state; low persistence therefore identifies rows that are poorly reconstructed by the dominant learned conditional organization.

At depth d, fit a conditional model $M_d = M(R_d)$ and robustly standardize lack of persistence,

$$Z_d^{\text{lack}}(x) = \frac{c_d - P_{M_d}(x)}{s_d}, \qquad c_d = \text{median} P_{M_d}, \tag{4}$$

$$s_d = \kappa_{\text{MAD}}\, \text{median}|P_{M_d} - c_d|. \tag{5}$$

Here $\kappa_{\text{MAD}} = \{\Phi^{-1}(3/4)\}^{-1} \approx 1.4826$ is the standard Gaussian consistency factor for the median absolute deviation. The next residual is

$$R_{d+1} = \{x \in R_d : Z_d^{\text{lack}}(x) > z_*\}, \qquad z_* = 0.5. \tag{6}$$

Peeling removes the statistically dominant core and refits the model on the surviving low-persistence population, allowing deeper heterogeneity or organization that is masked at the population level to become visible. Iteration yields $R_0 \supset R_1 \supset$

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TABLE I
DRIFT METRICS AND INTERPRETATIONS.

Coordinate	Name	Interpretation
Δ_C	Structural drift	Change in MAP-normalized masked-state reconstruction support after refitting.
Δ_K^{pseudo}	Information drift	Change in description length under the conditional structure learned from the data.
Δ_K^{gzip}	Kolmogorov drift	Compression-calibrated information displacement between adjacent residual levels.

$\cdots \supset R_D$ with a freshly fitted conditional model at every level; depth is structural, not chronological.

THREE SCALAR PROBES AT EVERY DEPTH

No single scalar probe captures all ways in which organization can change. We therefore retain three deliberately nonredundant views of each structural transition: Δ_C asks whether MAP-normalized reconstruction changes after refitting; Δ_K^{pseudo} asks whether the residual population has different conditional description length under the parent organization; and Δ_K^{gzip} asks whether adjacent residual populations are distinguishable through compression without using the fitted conditional model as the measuring instrument. These quantities need not move together—for example, Δ_K^{pseudo} can be nonzero even when $\Delta_C = 0$—so collapsing them to one coordinate would discard distinguishable forms of reorganization.

Structural drift. For observed state x_i, define

$$u_M(x,i) = \frac{\phi_i(x_i \mid x_{-i})}{\max_{a\in\mathcal{A}_i} \phi_i(a \mid x_{-i})}. \tag{7}$$

This is a MAP-normalized masked-state reconstruction score: $u_M = 1$ when the observed state attains the conditional MAP support, and smaller values measure its support relative to that MAP state. With a balanced probe cloud Q_d and common evaluable targets $\mathcal{I}_d(x)$,

$$\Delta_C^{(d)} = \frac{1}{|Q_d|} \sum_{x\in Q_d} \frac{1}{|\mathcal{I}_d(x)|} \sum_{i\in\mathcal{I}_d(x)} |u_{M_d}(x,i) - u_{M_{d+1}}(x,i)|. \tag{8}$$

Thus Δ_C measures how the system's learned ability to reconstruct masked coordinates changes across structural depth.

Information drift. The same learned conditional structure defines a data-adaptive code,

$$\ell_M(x) = -\frac{1}{n_M(x)} \sum_{i\in\mathcal{I}_M(x)} \log_2[\max\{\phi_i(x_i \mid x_{-i}), \varepsilon\}], \tag{9}$$

and

$$\Delta_K^{\text{pseudo},(d)} = \mathbb{E}_{R_{d+1}} \ell_{M_d} - \mathbb{E}_{R_d} \ell_{M_d}. \tag{10}$$

It is therefore a conditional-model-based description-length probe: information displacement is measured using structure learned from the system itself.

Kolmogorov drift. At depth d, draw matched row blocks $A_1, A_2 \subset R_d$ and $B_1, B_2 \subset R_{d+1}$. For serialized blocks a, b and compressor length $C(\cdot)$, normalized compression distance is[11]

$$\text{CD}(a,b) = \frac{C(a\|b) - \min\{C(a), C(b)\}}{\max\{C(a), C(b)\}}. \tag{11}$$

Within-depth distances establish the intrinsic compression scale, and the across-depth distance measures separation between adjacent levels:

$$\Delta_K^{\text{gzip},(d)} = \mathbb{E}\left[\text{CD}(A_1,B_1) - \tfrac{1}{2}\{\text{CD}(A_1,A_2) + \text{CD}(B_1,B_2)\}\right]. \tag{12}$$

At every depth d, the three numbers form

$$\mathcal{D}_X(d) = \left(\Delta_C^{(d)}, \Delta_K^{\text{pseudo},(d)}, \Delta_K^{\text{gzip},(d)}\right), \tag{13}$$

and stacking them gives the characteristic drift spectrum as a depth-by-coordinate matrix,

$$M_X = \begin{bmatrix} \mathcal{D}_X(0) \\ \mathcal{D}_X(1) \\ \vdots \\ \mathcal{D}_X(D-1) \end{bmatrix}. \tag{14}$$

GSS METHODOLOGY CHANGE: KNOWN DATA SHIFT

For the GSS results below, the target-wise conditionals are estimated using our LSM implementation; this is an instantiation of the conditional-learning objective above rather than a requirement of the spectrum. The GSS provides a particularly useful positive control because the pre-2021 waves define a tight empirical regime before a documented change in survey methodology. At the shallowest structural transition, the four waves from 2012–2018 cluster closely in the K–C plane, with $\Delta_K^{\text{pseudo}} = 0.0738, 0.0749, 0.0764$, and 0.0749 for 2012, 2014, 2016, and 2018, respectively (mean 0.0750). In 2021 this coordinate shifts abruptly to 0.1531, approximately twice the pre-2021 level. The subsequent mixed-mode waves partially return toward the earlier regime, with values 0.1420 in 2022 and 0.1117 in 2024. This temporal structure closely tracks the documented transition from predominantly face-to-face GSS collection through 2018, to primarily web-based collection in 2021, followed by mixed web/face-to-face bridge designs in 2022 and 2024[7–9]. GSS methodological studies report broad comparability of the later sequential mixed-mode designs while also documenting item-level mode effects[10]. The characteristic drift spectrum therefore recovers an independently known change in the observation process without being trained to discriminate survey years or modes.

A numerical representation makes the multiscale object explicit. Using the first three transitions and columns $(\Delta_C, \Delta_K^{\text{pseudo}}, \Delta_K^{\text{gzip}})$,

$$M_{2018} \approx \begin{bmatrix} .0971 & .0749 & .00061 \\ .1130 & .0460 & .00004 \\ .1164 & .0395 & -.00090 \end{bmatrix}, \tag{15}$$

$$M_{2024} \approx \begin{bmatrix} .1165 & .1117 & .00015 \\ .1383 & .0754 & -.00008 \\ .1472 & .0794 & -.00035 \end{bmatrix}. \tag{16}$$

These matrices are compact fingerprints of how the same three probes evolve with structural depth; comparisons between repeated observations can therefore use the full spectrum rather than a single classifier or one global distance.

INTERPRETATION AND NEXT STEPS

The characteristic drift spectrum is intended as a system-level measurement object whose primary distinction is between re-realization and reorganization. Its fundamental learning requirement is only a stable estimate of the target-wise conditional distributions that organize the multivariate state; it is not tied to a particular conditional-learning algorithm. For an

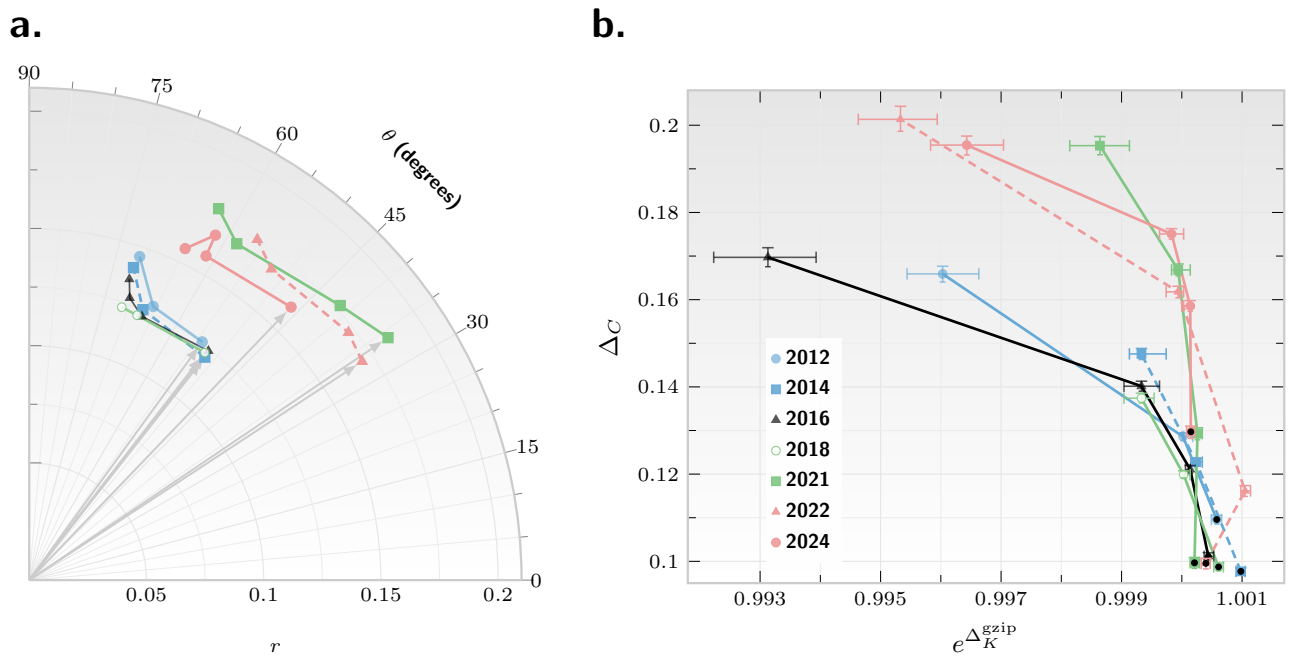


Fig. 1. Two projections of the GSS characteristic drift spectrum. Panel (a) shows a tight pre-2021 empirical regime across 2012–2018, followed by a pronounced 2021 displacement and partial return in 2022 and 2024, coincident with the documented change in GSS data-collection methodology from predominantly face-to-face collection through 2018, to primarily web-based collection in 2021, and then mixed web/face-to-face bridge designs in 2022 and 2024 [7–10]. The panel is a polar rendering of the raw $(\Delta_K^{\text{pseudo}}, \Delta_C)$ plane. Panel (b) shows gzip/NCD information displacement against the Dirichlet-posterior mean of Δ_C, with horizontal coordinate displayed as $e^{\Delta_K^{\text{gzip}}}$.

unchanged generating organization, repeated sufficiently large samples should approach the same spectrum; persistent spectral displacement instead indicates change in the structural or informational properties measured by the probes. The three coordinates are retained because they interrogate different, nonredundant manifestations of that change, while structural depth localizes where it occurs. The present results use LSMs as an available realization of the required conditional models, but alternative learners can be substituted without changing the definition of the spectrum. Matched-size refit nulls, independent peeling rules, planted-change simulations, alternative conditional learners, alternative compressors, and observation-design controls provide direct routes for quantifying finite-sample variability and calibrating the operating range of each coordinate.

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